Lesson 9

Topic: Understanding Context in DAX & CALCULATE and Basic Filters, Variables

1. What is row context? Give an example in a calculated column.

- **Row context** means DAX evaluates an expression *row by row* in a table.
- Example in a calculated column in **Sales**:
  Sales[TotalPrice] = Sales[Quantity] * Sales[UnitPrice]

👉 Here, for each row, DAX multiplies **that row's** Quantity × UnitPrice. That's row context in action.

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2. Write a measure that finds total sales

Total Sales = SUM (Sales[SalesAmount])

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3. Use RELATED to fetch the Name from the Customers table into the Sales table

Sales[CustomerName] = RELATED (Customers[Name])

👉 Works only if a relationship exists from Sales → Customers.

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4. What does this return?

CALCULATE (SUM (Sales[Quantity]), Sales[Category] = "Electronics")

✅ Returns the **total quantity sold** for only **Electronics** category (ignoring other categories, but still respecting slicers like Date, Region, etc.).

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5. Difference between VAR and RETURN

- VAR stores an intermediate value (scalar, table, or expression).
- RETURN tells DAX what final value to output.
  Example:

```
Profit Margin =
VAR Profit = SUM ( Sales[SalesAmount] ) - SUM ( Sales[Cost] )
VAR SalesTotal = SUM ( Sales[SalesAmount] )
RETURN DIVIDE ( Profit, SalesTotal )
```

---

6. Create a calculated column in Sales called TotalPrice using row context

Sales[TotalPrice] = Sales[Quantity] * Sales[UnitPrice]

---

7. Write a measure Electronics Sales using CALCULATE

```
Electronics Sales =
CALCULATE (
   SUM ( Sales[SalesAmount] ),
   Sales[Category] = "Electronics"
)
```

---

8. Use ALL(Sales[Category]) to ignore category filters

```
Total Sales (Ignore Category) =
CALCULATE (
    SUM ( Sales[SalesAmount] ),
    ALL ( Sales[Category] )
)
```

---

9. Fix this error: RELATED(Customers[Region]) returns blanks

👉 Causes:

- No relationship between Sales and Customers.
- Or relationship is inactive.

✅ Fix: Create/activate a relationship between Sales[CustomerID] and Customers[CustomerID].

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10. Why does CALCULATE override existing filters?

- Because CALCULATE **modifies filter context**.
- When you pass conditions (Sales[Category] = "Electronics"), it **replaces** the existing filter on that column.

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11. Write a measure that returns average unit price

```
Average Unit Price = AVERAGE ( Sales[UnitPrice] )
```

---

12. Use VAR to store a temporary table of high-quantity sales (Quantity > 2), then count rows

```
High Quantity Count =
VAR HighQ =
    FILTER ( Sales, Sales[Quantity] > 2 )
RETURN COUNTROWS ( HighQ )
```

---

13. Measure: % of Category Sales

```
% of Category Sales =
DIVIDE (
    SUM ( Sales[Sales] ),
    CALCULATE (
        SUM ( Sales[Sales] ),
        ALLEXCEPT ( Sales, Sales[Category] )
    )
)
```

👉 Each sale’s contribution relative to its category.

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14. Simulate a "remove filters" button using ALL

```
Sales (Ignore Region) =
CALCULATE (
    SUM ( Sales[Sales] ),
    ALL ( Sales[Region] )
```

)
👉 When users click this measure (e.g., in a card), it always ignores Category slicers.

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15. Troubleshoot: A CALCULATE measure ignores a slicer. Likely cause?

- The slicer is on a **different table** with **no relationship** to Sales.
- Or ALL() is being used inside the measure, which removes slicer filters. ✅ Fix: Check model relationships, and remove unnecessary ALL() if you want slicers to apply.